

Blockchain Oracles

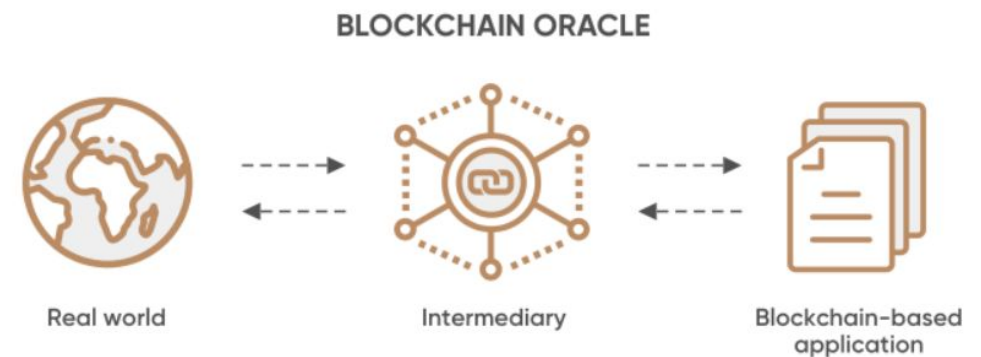
Chapter – 6

Fall 2025

Middle Tennessee State University

Introduction to Blockchain Oracles

- Entities that connect blockchains to external systems
- Third-party service that connects smart contracts with the outside world
- Serve as bridges connecting the blockchain world with the real world
- Enables smart contracts to execute based upon inputs and outputs from the real world
- Enables smart contracts to access and respond to external data
- Expand the capabilities of smart contracts

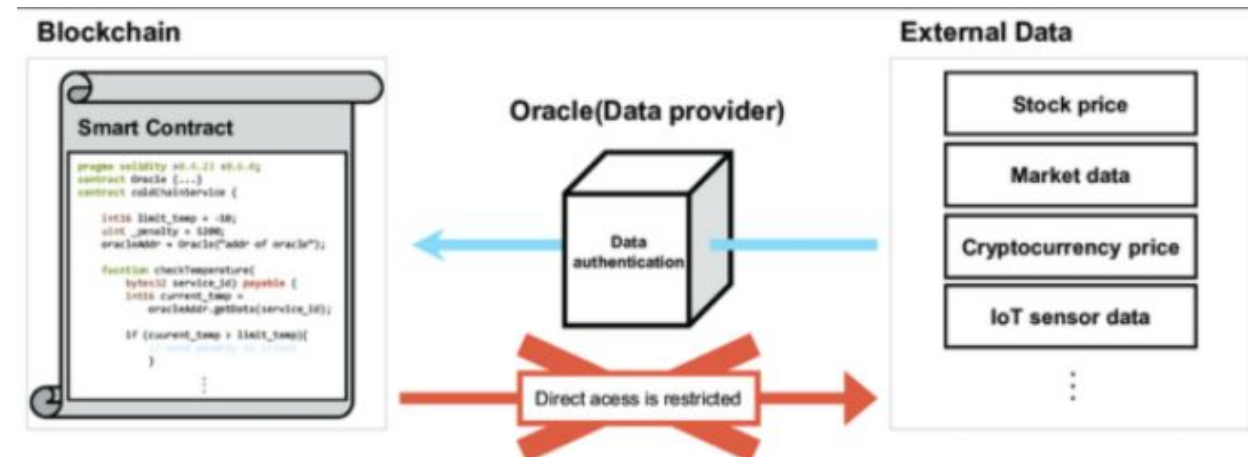


Introduction to Blockchain Oracles (Contd...)

- Makes a smart contract more versatile and practical for a wide range of applications
- Example 1:
 - Decentralized Finance (DeFi) platforms rely on oracles for real-time price feeds of cryptocurrencies and other assets
- Example 2:
 - An insurance contract automatically triggers a payout to farmers when an oracle reports drought conditions meeting predefined criteria
- Example 3:
 - Supply chain platforms employ oracles to verify the status and location of products in real time.

The need for Blockchain Oracles

- Blockchains are usually “**isolated**”
- Key limitation of Smart Contract: **cannot connect** with data and systems outside of their native blockchain
- External resources are referred to as “**off-chain**”
- data currently recorded on the blockchain is referred to as “**on-chain**”
- **Oracle**: bridge the gap between “off-chain” and “on-chain”
- Blockchains have limited access to real-world data
 - making external data crucial for many use cases
 - This is why Oracles are essential
- Use Cases in Decentralized Finance (DeFi), Supply Chain, Insurance, etc.
 - DeFi platforms, supply chain tracking, and insurance rely on oracles for real-world data.



How do Blockchain Oracles work?

- **Step 1: Data Request by Smart Contract**

- *E.g., A DeFi lending smart contract needs real-time price data for a cryptocurrency as collateral. It initiates a request for the current price of Bitcoin.*

- **Step 2: Oracle Selection**

- *E.g., The DeFi platform selects a decentralized oracle network like Chainlink to fetch the data due to its reputation for reliable data feeds.*

- **Step 3: Data Source Access**

- *E.g., The Chainlink oracle accesses multiple cryptocurrency exchanges to collect Bitcoin price data.*

- **Step 4: Data Aggregation and Verification**

- *E.g., Chainlink aggregates price data from multiple exchanges and verifies it to ensure its accuracy.*

- **Step 5: Data Reporting to Smart Contract**

- *E.g., Chainlink reports the current Bitcoin price to the DeFi smart contract, which is in the form of a blockchain transaction.*

- **Step 6: Smart Contract Execution**

- *E.g., The DeFi smart contract evaluates the Bitcoin price against the collateral requirements and automatically executes actions like issuing a loan or triggering a liquidation if necessary.*

- **Step 7: Result Verification**

- *E.g., The blockchain network validates the execution of the DeFi smart contract to ensure that it followed the agreed-upon rules and that the reported data was accurate.*

- **Step 8: Reward Distribution**

- *E.g., Chainlink nodes and the DeFi protocol might receive tokens as rewards for their services, encouraging data accuracy and network participation.*

- **Step 9: Data Update and Monitoring**

- *E.g., The Chainlink oracle continues to monitor Bitcoin price data and provides real-time updates to the smart contract as needed.*

- **Step 10: Data Management**

- *E.g., Historical data, including past Bitcoin prices, may be stored on-chain for auditing and transparency.*

- **Step 11: Security Measures**

- *E.g., Data encryption and digital signatures are used to ensure data integrity and prevent unauthorized access or manipulation.*

Types of Blockchain Oracles

1. Software Oracles

- software programs that retrieve data from external sources

Strength:

- fetch data from a wide range of sources, including APIs, websites, and databases
- provide real-time data, making them suitable for time-sensitive applications
- often cost-effective to implement

Weakness:

- Verification of the reliability of data providers
- Software oracle relies on a centralized source

Example:

- A DeFi lending platform uses a software oracle to access cryptocurrency price data, enabling automated loan-to-value ratio calculations and collateral valuation

Cryptocurrency Price Oracles

Weather Data Oracles

Sports Data Oracles

Stock Market Price Oracles

Flight Information Price Oracles

Types of Blockchain Oracles (Contd...)

2. Hardware Oracles

- physical devices or sensors that connect real-world data to the blockchain

Strength:

- connect blockchain to physical devices, such as IoT sensors, ensuring reliable real-world data inputs
- tamper-resistant and can provide higher security for data inputs.

Weakness:

- Developing and deploying hardware oracles can be more complex and expensive
- best suited for applications where physical data is essential

Example:

- An agricultural blockchain uses IoT sensor-based hardware oracles to monitor soil conditions, ensuring accurate data for crop management and supply chain tracking.

IOT Sensors

RFID Tags

Smart Meters

Heart Rate
Monitors

Geolocation
Devices

Types of Blockchain Oracles (Contd...)

3. Inbound Oracles

- responsible for fetching external data and bringing it onto the blockchain

Strength:

- enable the blockchain to access and use real-world data for various applications
- Can verify and validate data from external sources, enhancing reliability.
- Support a wide range of data sources and use cases

Weakness:

- Still dependent on external data sources for accuracy
- If data sources are centralized, it can introduce centralization risks

Example:

- An inbound oracle retrieves real-time stock market data from multiple sources, allowing decentralized applications to make investment decisions based on accurate market information.

Types of Blockchain Oracles (Contd...)

4. Outbound Oracles

- facilitate the transmission of data or instructions from the blockchain to external systems or devices

Strength:

- Enable smart contracts to trigger real-world actions or payments
- Can interact with IoT devices, payment gateways, and other external systems.

Weakness:

- Relies on external systems to execute actions

Example:

- An outbound oracle linked to a smart lock system allows a blockchain-based access control smart contract to automatically unlock a door when a payment is received.

Types of Blockchain Oracles (Contd...)

5. Centralized Oracles

- rely on a single, centralized data source for providing information to smart contracts

Strength:

- Easier to implement as they depend on a single data source
- Often more cost-effective for applications with limited data needs

Weakness:

- Vulnerable to data manipulation and disruptions from a single source
- Trust dependency as smart contracts rely heavily on the accuracy and reliability of the centralized source.

Example:

- A centralized price oracle for a *stablecoin* relies on a specific exchange to provide cryptocurrency price data. If the exchange experiences downtime or provides incorrect data, it can negatively impact the entire ecosystem.

Types of Blockchain Oracles (Contd...)

6. Decentralized Oracles

- aggregate data from multiple sources and use a network of nodes to validate and verify the data.

Strength:

- Reduced risk of incorrect information due to data aggregation and verification.
- Enhanced security against data manipulation and centralized control
- Maintain trustlessness and decentralization principles of blockchain

Weakness:

- Setting up and managing a decentralized oracle network can be more complex.
- consensus process can introduce delays in reporting.
- Example:
 - Chainlink, a leading decentralized oracle network, aggregates data from various cryptocurrency exchanges and relies on a network of nodes to verify and provide accurate data for DeFi applications.

Types of Blockchain Oracles (Contd...)

7. Human Oracles

- rely on human input or decision-making processes to provide data or determine outcomes in smart contracts

Strength:

- Diverse data sources as multiple individuals contribute real-world information, increasing data reliability
- Less susceptible to manipulation due to the collective wisdom of the crowd
- Humans can provide data on events that are not easily quantifiable by machines

Weakness:

- Dependence on human input can lead to slower response times for oracle data.
- Inaccurate reporting may occur due to mistakes or manipulative reporting.
- Challenges in aligning the economic interests of human reporters with accurate reporting.

Example:

- Augur, a decentralized prediction market platform, relies on human oracles to report the outcomes of events like sports scores, election results, or even weather conditions, based on crowd consensus.

Benefits of Blockchain Oracles

- **Data Integration:** Oracles empower blockchain networks with **real-world information**, broadening their use cases
 - *Example: In the world of decentralized finance (DeFi), lending platforms rely on oracles to provide accurate and up-to-date price feeds of cryptocurrencies and other assets. This integration enables automatic collateral valuation, loan-to-value ratio calculations, and secure lending services.*
- **Automation:** Oracles enable the trustless and **automated execution of smart contracts** based on real-world events.
 - *Example: Consider an insurance contract that utilizes an oracle to access real-time weather data. When predefined criteria for extreme weather conditions are met, the smart contract automatically triggers insurance payouts to farmers, eliminating the need for manual claims processing and reducing delays.*
- **Decentralization:** Well-designed oracles can maintain the **decentralized nature** of blockchain platforms.
 - *Example: Chainlink, one of the leading decentralized oracle networks, aggregates data from multiple sources and employs a network of node operators to ensure data integrity. This approach aligns with the decentralized principles of blockchain technology and provides reliable and tamper-resistant data.*
- **Versatility:** Oracles can provide data on a **wide range of real-world events**, making them suitable for various applications.
 - *Example: Oracles are used to verify the location and status of products in a global supply chain. An oracle confirms the successful delivery of a shipment, which, in turn, triggers the release of payment to the supplier. This process minimizes disputes, enhances supply chain efficiency, and ensures timely transactions.*
- **Real-time Data:** Oracles provide access to real-time data, which is essential for **time-sensitive applications**.
 - *Example: In the context of sports betting and gaming, oracles can deliver real-time scores and game results to smart contracts. This ensures instant settlement of bets and a seamless user experience.*

Challenges and Risks

- **Data Reliability:** Relying on external data sources introduces a **level of trust in the accuracy of that data**.
 - *Example: A decentralized lending platform uses a price feed oracle. If the oracle relies on a single, unverified data source, it may expose the platform to incorrect or manipulated data, leading to incorrect liquidations or lending decisions.*
- **Data Manipulation:** Malicious oracles or compromised data sources **may provide false data, leading to vulnerabilities** in smart contracts
 - *Example: In a decentralized insurance application, if a weather oracle is manipulated to report false weather conditions, it could trigger unwarranted insurance payouts, resulting in financial losses.*
- **Centralization Risk:** Centralized oracles can pose central points of failure and **data manipulation risks**.
 - *Example: A centralized oracle service used by a DeFi protocol experiences downtime due to a technical issue or external pressure, disrupting the functioning of numerous smart contracts that depend on its data.*
- **Privacy Concerns:** Oracles may access **sensitive data**, raising concerns about data privacy
 - *Example: An oracle that retrieves health data for a blockchain-based medical record system must ensure the privacy and security of the patient's information to comply with regulations like HIPAA.*
- **Oracle Dependence:** Smart contracts relying heavily on oracles may become vulnerable if the **oracle service fails or becomes unavailable**.
 - *Example: An auction platform that depends on an oracle for real-time bidding data may experience a disruption in the auction if the oracle goes down, causing potential losses.*
- **Accuracy Verification:** Ensuring the accuracy of oracle data requires **additional processes and costs**.
 - *Example: To guarantee the accuracy of a financial data oracle, a blockchain project may need to implement mechanisms for cross-referencing data from multiple sources, incurring additional operational costs.*

Future Trends and Considerations in Oracles

- **Cross-Chain Oracles**

- development of oracles capable of bridging data across multiple blockchains, **enabling interoperability** between different blockchain networks
- *Example: A cross-chain oracle solution allows assets to move seamlessly between Ethereum and Polkadot, facilitating decentralized applications that span multiple blockchains*

- **Privacy-Focused Oracles**

- Solutions that enable fetching sensitive data from external sources while **preserving the privacy of that data**.
- *Example: An oracle that retrieves private financial data from a bank's records for blockchain-based auditing without exposing sensitive customer information.*

- **Hybrid Approaches**

- Combining decentralized and centralized elements to **balance data verification and reliability**.
- *Example: A supply chain oracle network that utilizes a combination of decentralized node operators and a centralized verification authority to ensure both data reliability and efficiency in tracking products.*

- **Innovations in Data Sources**

- Integration with emerging data sources, **including IoT devices, real-time sensors, and data generated by smart cities**.
- *Example: An IoT-based oracle network retrieves real-time environmental data from smart city sensors to provide information for urban planning and environmental monitoring smart contracts.*

- **Cross-Industry Integration**

- Expanding the adoption of oracles beyond financial services to industries like **healthcare, logistics, and energy**.
- *Example: An oracle network is used in the healthcare sector to enable secure and verified sharing of patient medical data for research purposes while maintaining patient privacy.*

Notable Oracle Projects

- **Chainlink (LINK)**
 - *Example: Chainlink provides price feeds for various cryptocurrencies and assets, supporting DeFi applications and decentralized exchanges.*
- **Band Protocol (BAND)**
 - *Example: Band Protocol aggregates data from multiple sources to provide accurate and reliable information for DeFi and smart contract applications.*
- **Augur (REP)**
 - *Example: Augur's users report outcomes of events like sports scores and election results, contributing to the accuracy and trustworthiness of the platform.*
- **DIA (DIA)**
 - *Example: DIA's oracles supply data for various DeFi platforms, including lending protocols and decentralized exchanges.*



Reminder/Announcement

- Project Progress Report (Phase 2) – Deadline extended to Thursday, 11:59 PM
- **Results and Discussion – Performance Evaluation (Phase 3)**
 - Metrics and Results that is based on YOUR project
 - **Metrics/Results:** Performance testing of your smart contract in **Sepolia Testnet Blockchain** when you feed your 'input dataset' that you have collected or created
 - You need to provide the reference link in case you have downloaded the input dataset from Internet that you feed to your smart contract
 - You will get data for results from Sepolia Testnet Blockchain (<https://sepolia.etherscan.io/>) after you deploy your smart contract in Sepolia and provide inputs (interact) with your smart contract to create transactions
 - Or, just follow the **automation contract caller** tutorial I sent you that will collect all results in a JSON file.
 - Representation of your performance evaluation/metrics/results can be in the form of:
 - Graphs and Plots (ensure you have identified the X and Y-axis units)
 - Pie-charts
 - Tables
 - Etc.

Reminder/Announcement – Contd...

- Some examples of metrics/results for performance evaluation of your project are:

[Note: Results should be based on Sepolia Tesnet Blockchain]

- ❖ Evaluation of the Transaction Costs and Gas Costs
 - ❖ Average of different Transaction Costs obtained from different transactions
 - ❖ Evaluation of the Transaction Time
 - ❖ Average of different Transaction Time obtained from different transactions
 - ❖ Evaluation of which function out of all from your smart contract consumed more gas and txn cost
 - ❖ Comparison of all your functions in terms of transaction/gas costs.
 - ❖ What kind of curve distribution your results have? Gamma?, Normal?, Binomial?, Exponential?
Based on the kind of curve you get. And the curve is based on the result you get
 - ❖ **Only gas cost or time not enough; application-specific metrics is a must**
Think of more metrics for performance evaluation and comparison. The more the better the evaluation!
-
- ✓ **Write at least 5-8 lines describing the significance and impact of each result**
 - ✓ **How do independent and dependent variables vary with respect to control parameters?**
 - ✓ **Linear/non-linear behavior? If so, why?**

Reminder/Announcement – Contd...

Submission Deadline: Tuesday, November 11, 11:59 PM

1. Submission of Word/pdf document – Just like previous project drafts submissions
2. Attachments of your code, dataset for input, and relevant CSV files (Created from JSON file from automated contract caller script OR, from Sepolia Testnet)
3. **The submission of word/pdf should have the following:**
 - **Provide URL link of the dataset that you downloaded from Internet – dataset you use to feed to your smart contract as input data. The instructor should be able to access this internet available dataset that you're using**
 - 5-8 lines of explanation of the dataset
 - Why did you decide to download this particular dataset?
 - How many rows and columns? Any cleaning and preprocessing of the data needed?
 - What variables/features from dataset you decided to choose to input for your smart contract?
 - **Observation and explanation (5-8 lines) of your results: plots, graphs, curves, charts, tables, etc.**
 - **Write at least 5-8 lines describing the behavior, significance and impact of each results that you get**
 - **Refer to this paper for an example:**
https://engineering.unt.edu/cse/research/labs/nsi/sites/default/files/biblio/documents/digital_ledger_spammer_resume.pdf

End of Chapter-6